NOTES — Data quality, inference limits and production design

Scope and evidence. The local implementation processes 1,500 synthetic customers (July–September 2026). Observed evidence: **135,155 source rows = 133,692 active clean rows + 1,463 rejected/superseded rows**; 54/54 recorded local tests pass; all four required CSVs were byte-identical on rerun. The fresh three-file run took **7.678 s**, peak **82.8 MiB RSS** (one workstation, not a scale benchmark). The production designs and monitoring gates below are **proposals**, not deployed controls.

1. What EDA revealed — and what we changed

Raw-data finding	Pipeline decision / data-quality control
July/August CSV (`transaction_id`, ISO date, signed `amount`, `currency`) becomes September JSONL (`txn_id`, `posted` in `DD/MM/YYYY`, positive `value`, `direction`, `narrative`); 22,476 September values contain `£`.	Versioned schema adapters first; normalise dates, GBP magnitudes and debit/credit signs into a canonical Silver contract. Never infer monetary sign from `value` alone.
1,423 repeat-ID occurrences (including 312 September IDs with changed payloads); 40 July rows without `customer_id`.	Bank-scoped stable transaction key; payload fingerprint; reject missing identity; preserve duplicate/superseded versions and reasons in audit. Deterministic **latest-valid-observed** precedence is a convention, **not** proof of banking-system authority. Never deduplicate solely on description/date/amount.
1,052 July-booked transactions arrive in August; 184 raw merchant strings collapse to 92 after case normalisation.	Track *delivery period separately from booking period*; support backfills and changed monthly feature windows. Canonicalise merchant matching without overwriting original descriptions.

At Bronze ingestion, verify manifest checksum and row count, required fields, parseability, permitted currency/direction, identity completeness, and reconciliation. Fail/quarantine unsupported schema versions rather than silently discard fields. Silver emits canonical records and an explicit rejected/superseded audit; Gold bill/switch outputs are derived reproducibly. An identical input manifest is a no-op; a changed payload is versioned and downstream affected windows are recomputed.

2. Why our inference is intentionally conservative

Label sufficiency is the binding constraint. The 57 original manually labelled descriptors are imbalanced across 10 target classes (only **one gym** and **two home-insurance** examples) and include incorrect labels (`OCTOPUS ENERGY` and `TESCO MOBILE`). Repeated transactions do not create independent labels. A complex *newly trained* supervised model (e.g., XGBoost) would be particularly vulnerable to overfitting and falsely confident minority-class assignments. We did **not** reject machine learning: character TF-IDF similarity is our lightweight statistical retrieval model; pretrained, offline MiniLM is diagnostic only.

Decision funnel: approved merchant alias (versioned, reviewed) → lexical nearest-label candidate → MiniLM agreement/disagreement as **non-voting evidence** → accept only under approved policy, otherwise a valid 10-class provisional category plus `needs_review=true`. Review-required predictions are withheld from trusted recurring-bill/switch inference. This hybrid is more **auditable and operationally safe**, not proven more accurate than every alternative classifier. Similarity is not calibrated probability; no network inference is needed.

Leave-one-description-out exploratory TF-IDF accuracy **0.789**, macro-F1 **0.635**, macro-average-precision **0.698** on just 57 labels (MiniLM: **0.754 / 0.657 / 0.654**). These are **not** independent production accuracy estimates. Current accepted transaction coverage is **88.76%**; **21/92** descriptions (15,033 transactions; **11.24%**) require review. Accepted does **not** mean verified correct: two accepted Admiral aliases (851 transactions) carry high-risk audit flags, requiring governance resolution before production. Better algorithms cannot infer merchant ground truth absent informative labels or identities; expanding approval indiscriminately merely conceals ambiguity.

What else we do not trust: 6,681 recurring bills and 209 switches are *candidates*, not confirmed contracts/cancellations. Of the switches, 87 are provisional, 122 show repeat new-provider evidence, and **none is externally confirmed**. Observed lower payments cannot establish causal or annual savings; the report must retain category-level uncertainty and avoid marketing those estimates as realised savings.

3. Monthly production: two deployment boundaries

A — ApTap-managed AWS, only where transfer is authorised: Bank-approved encrypted, resumable export + checksummed READY manifest → private S3 immutable Bronze → EventBridge/Step Functions → Glue/EMR Spark validation/quarantine → Iceberg Silver `MERGE` and correction audit → incremental customer-period Gold → pinned **offline** classifier/review registry → reports. Use IAM least privilege, KMS, private networking, retention/deletion controls, audit logs, and data-quality/cost alerts. Unfinished multipart deliveries are not promoted as complete batches.

B — Fully bank-contained (15m customers; no data egress): Bank source → approved resumable/chunked internal handoff, per-chunk/full hash and atomic manifest → bank-native object storage → approved Spark/SQL scheduler + Bronze/Silver/Gold → signed, pinned offline classifier *or approved native equivalent* → bank-local review and dashboard. No unapproved package installs, external model calls, telemetry exports, or model-weight downloads. Bank change control governs source code, dependencies/SBOM, alias policies, model versions, testing and rollback; only explicitly approved de-identified aggregates may leave, if permitted at all.

Scale/storage/cost. September had 44,950 rows/1,500 customers: at the same activity, **15m customers 449.5m rows/month**, approximately **64.6 GB raw JSONL** at September's density, *before* replicas, Parquet, indexes and audit. Naively projecting the measured local runtime gives **~7.1 h/month**, **not** a distributed SLA or valid linear memory forecast. SQLite's single writer, global ID lookups, full-history recompute, skew and failed large-file transfers break first. Partition by bank and booking month, preserve source-delivery partition/lineage, cluster by customer/date, maintain an indexed bank+transaction-ID lookup, compact small files, and recompute changed windows only. Transaction IDs, customer references, merchant text, dates, amounts and spending patterns are sensitive; minimise access, log no raw payload/identifiers and apply contract-specific retention. **Illustrative compute-only cost:** 16 Glue DPUs × 4 h × assumed \$0.44/DPU-h = **\$28.16 / 449.5m \$0.063 per million transactions**; no throughput or price guarantee, excluding storage, transfers, security, orchestration, inference and human review. Measure all-in actual cost/million in a load test.

4. MLOps: benchmark, drift, response and release

Keep **macro-F1** as the primary *labelled* ten-class score so minority categories contribute equally; inspect per-class precision/recall and one-vs-rest average precision (PR) as diagnostics. Benchmark a majority-class predictor and exact-approved-alias rules against the frozen TF-IDF champion and optional MiniLM challenger. The **0.635 exploratory macro-F1 is not a production acceptance baseline**. Establish a versioned, independently adjudicated merchant-level reference set, with stratified rare-class/new-merchant sampling, hidden holdout and uncertainty intervals; pool months when a class has too few reviewed examples. Separately record approved coverage, review rate, risk overrides, missingness, reconciliation, latency and cost.

Proposed gate: investigate when *observed* monthly macro-F1 falls by **at least 10% relative** to the validated champion baseline (for example **0.70 → 0.63**) on adequately labelled comparable samples; confirm across two windows or a material uncertainty-adjusted decline. Initiate **review/retraining evaluation**, not automatic model promotion. Independently monitor PSI on stable, binned input features (amount bands, text length, known/unseen merchant share) and output category mix: provisional warning **PSI of 0.10 or higher**, investigation **0.20 or higher**; calibrate for seasonality and bank-specific distributions. PSI detects **distribution shift, not degraded prediction accuracy**. Prioritise human labels and registry corrections; train a challenger only with sufficient new ground truth, test against untouched holdout, check minority-class and high-risk errors, then release/rollback through the relevant approval process.

First week: jointly confirm contracts, bank security and lateness/correction semantics; run October-style schema/backfill and 15m-scale synthetic load tests; implement batch and data-quality alerts and offline review workflow; collect an independently adjudicated validation sample; baseline runtime, drift, macro-F1 and *all-in* cost before agreeing an SLA.

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